

ZAINAB IRFAN

AI/ML Engineer • RAG & LLM Systems Specialist • Full-Stack Python Developer

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PROFESSIONAL PROFILE

Results-driven AI/ML Engineer with hands-on, production-grade experience designing and shipping **Retrieval-Augmented Generation (RAG)** systems, conversational **LLM** agents, and full-stack AI platforms. Proven track record delivering end-to-end solutions — from data pipelines and model integration to deployed, customer-facing applications — using **FastAPI**, **Next.js**, **LangChain**, and vector databases such as **Qdrant**. Combines strong software engineering fundamentals with applied machine learning expertise, backed by 7 industry certifications from **AWS**, **IBM**, and DeepLearning.AI. Adept at translating business requirements into scalable, reliable AI products and comfortable owning projects independently in fast-paced, client-facing environments.

CORE TECHNICAL EXPERTISE

AI / LLM Engineering — RAG Pipeline Architecture · LangChain · Prompt Engineering · LLM Evaluation & Benchmarking · Conversational AI Agents · Streaming Chat Interfaces · Workflow Automation (n8n)

Machine Learning — Supervised & Unsupervised Learning · Feature Engineering · Model Evaluation · Bias Detection · Exploratory Data Analysis (EDA)

Languages & Frameworks — Python · JavaScript · HTML/CSS · FastAPI · Streamlit · Next.js · React

Data & Vector Infrastructure — Qdrant · Supabase · NumPy · Pandas · Scikit-learn · Matplotlib · Seaborn

APIs, Cloud & Deployment — REST API Design & Integration · AWS (AI/ML) · Render Deployment · Infrastructure Automation

Languages Spoken — English (Proficient) · Urdu (Native)

WORK EXPERIENCE

Freelance AI/ML Engineer & Full-Stack Developer 2024 – Present

Self-Employed / Independent Contractor

- Designed and built multiple production-style **RAG** platforms and **LLM**-powered applications, including document intelligence systems, conversational support agents, and workflow automation tools, using **FastAPI**, **Next.js**, **LangChain**, and **Qdrant/Supabase**.
- Owned the full software development lifecycle independently — API design, data pipeline development, model integration, cloud deployment (**Render/AWS**), and post-launch iteration — while managing client requirements and delivery timelines.
- Applied machine learning and data analysis techniques (**EDA**, feature engineering, model evaluation) to deliver predictive insights for clients across e-commerce and clinical/healthcare data use cases.
- Built automation systems (e.g., **LLM**-driven email response pipelines with **n8n**) that reduced manual operational workload for client businesses.

KEY PROJECTS

Prism AI — RAG-Based Document Intelligence Platform 2026

Stack: Next.js, FastAPI, LangChain, Qdrant, Supabase, OpenAI/Claude APIs · Deployed on Render

- Built a full-stack **Retrieval-Augmented Generation** platform that ingests multi-format documents (**PDF**, **DOCX**, **PPT**, **CSV**) and returns structured, context-aware answers.
- Designed the retrieval pipeline with **LangChain** and **Qdrant** for semantic search, using **Supabase** for persistent storage of documents and chat history.
- Integrated **LLM APIs** (**OpenAI** and **Claude**) with streamed chat responses for a real-time conversational Q&A experience.
- Built an analytics dashboard visualizing usage insights, queries per day, question volume, and engagement trends, with interactive charts.
- Deployed a production build with a **Next.js** frontend and a **FastAPI** backend hosted on **Render**.

Skills Applied: RAG Pipeline Design, LangChain, Vector Search (Qdrant), FastAPI, Next.js, LLM API Integration, Streaming Interfaces, Cloud Deployment, Data Visualization

AI Customer Support Agent for Shopify Stores2025 – 2026

Stack: Python, LLM APIs, Shopify Integration

- Developed an AI-powered customer support agent that integrates with a **Shopify** storefront to handle customer queries automatically.
- Used **LLM APIs** to generate context-aware responses, reducing manual reply workload for store operations.

Skills Applied: LLM API Integration, Conversational AI Agents, Third-Party API Integration, Python

Automated Email Reply System2025

Stack: Python, LLM APIs, Email Automation, n8n

- Built an automation system that reads incoming emails, generates context-appropriate draft replies using an **LLM**, and sends responses automatically.

Skills Applied: Workflow Automation (n8n), LLM Prompt Engineering, Python, Email/API Integration

Comprehensive Clinical Data Analysis & Visualization2025

Stack: Python, Pandas, Matplotlib/Seaborn, Scikit-learn

- Performed end-to-end exploratory data analysis (**EDA**) on raw clinical datasets, including cleaning, transformation, and feature engineering.
- Built visualizations of clinical insights such as patient distribution by disease and other key trends across the dataset.
- Applied machine learning techniques to generate predictive insights from patient records.

Skills Applied: Exploratory Data Analysis, Feature Engineering, Pandas, Data Visualization (Matplotlib/Seaborn), Scikit-learn, Predictive Modeling

AI PDF Question-Answering Platform2025

Stack: Python, FastAPI, LLM APIs

- Built a tool that allows users to upload a **PDF** and ask natural-language questions, returning direct answers sourced from the document.

Skills Applied: FastAPI, Document Parsing, LLM API Integration, REST API Design

LLM Platform Evaluation & BenchmarkingIn Progress, 2026

Stack: Python, LLM APIs, Open-Source & Hugging Face Models, Model Integration

- Currently developing a benchmarking framework to evaluate and compare the performance of different **LLM** platforms across defined tasks and metrics.

Skills Applied: LLM Evaluation & Benchmarking, Hugging Face Models, Python, Model Integration

Machine Learning Models Development2024 – 2025

- Trained 5+ classification and regression models with a full preprocessing pipeline: encoding, normalization, scaling, and train/test splitting.
- Evaluated models using accuracy, F1-score, precision/recall, and confusion matrices; selected optimal algorithms per dataset characteristics.

Skills Applied: Supervised & Unsupervised Learning, Model Evaluation, Scikit-learn, Feature Engineering

RESTful API Development & ML Integration2024

- Engineered **FastAPI**-based **REST APIs (GET/POST/PUT/DELETE)** with request validation, integrating trained ML models as live prediction endpoints.

Skills Applied: FastAPI, REST API Design, ML Model Deployment, Request Validation

CERTIFICATIONS

- AI Fundamentals & Cloud — **AWS**
- AI Workflow: Feature Engineering & Bias Detection — **IBM**
- Building Systems with the ChatGPT API — **DeepLearning.AI**
- Machine Learning for All — **University of London**
- Machine Learning — **Simplilearn**
- App Deployment & Security — **Packt**
- Learn to Code with AI — **Scrimba**